1. Display Basics (Functions)

Display is used to **show information visually** (numbers, text, images).Functions of Displays

1. **Output the correct display code**
   - The system sends the correct code to show numbers or characters.
2. **Send the code to the correct display**
   - If there are many displays (for example 4 displays in a row), the system sends the code to the **correct position**.

Example:A row of **4 hexadecimal displays** may show numbers like **0–9 and A–F**.

Types

- **Non-multiplexed displays** → each display has its **own connection**.
- **Multiplexed displays** → many displays **share connections** and work **one by one very fast**.

2. Introduction to Display Devices

A **display device** is used to **show information visually**.

Examples:Computer monitor, TV screen

Display devices show:Text,Images,Videos

3. Types of Display Devices

1. Analog Display Devices

These use **Cathode Ray Tubes (CRT)**.

Examples:Oscilloscope,Old TV CRT screens

2. Digital Display Devices (Flat Panel)

Modern displays are digital.

Examples:

- **LED display**
- **OLED display**
- **LCD (Liquid Crystal Display)**

- **TFT-LCD**
- **Plasma Display Panel (PDP)**

3. Other Modern Displays

Examples:

- **Electronic paper (e-paper)** – used in e-book readers
- **Nanoelectronics displays** – use nanotechnology like **carbon nanotubes**

Attributes of a Good Display Device

These are the **important qualities of a good display**.

1. Response Time-The **time required for a pixel to change color or brightness**.**Lower response time = better display**.

2. Persistence-The **time a phosphor continues to glow after being hit by an electron beam**.**If persistence is low, the screen must refresh faster to avoid flicker.**

3. Aspect Ratio-Ratio of **vertical pixels to horizontal pixels**.**Example:Standard PC aspect ratio = 4 : 3If the ratio changes, the image becomes stretched or distorted.**

4. Brightness-The **amount of light produced by the display**.**More brightness → clearer image.**

Display Device Principles

Display devices can be two types:

Active Displays

- Produce light by themselves.Based on **luminescence**.Example:LED displays

Passive Displays

- **Reflect or control light** instead of producing it.**Example:LCD displays**

More Attributes of a Good Display

1. Pixel Resolution

- Number of **pixels in a display screen**.

Example (VGA):720 × 400 pixels (text mode),640 × 480 pixels (graphics mode)More pixels → **clearer image**.

2. Display Size

- Measured **diagonally from one corner to another**.Unit: inches.Example:15 inch monitor,24 inch monitor

3. Viewing Angle

- The **angle from which you can clearly see the screen from the side**.
- **CRT → larger viewing angle**
- **LCD → smaller viewing angle**

Important Terminology

1. Pixel

A **pixel (picture element)** is the **smallest unit of information in an image**.The screen contains **thousands of pixels arranged in rows and columns**.Many pixels together form an image.

2. Resolution-Resolution is the number of distinct pixels in each dimension that can bedisplayed. More the resolution more clear is the image .

3. Aspect Ratio

Aspect ratio = **X pixels : Y pixels**

Example:**4 : 3** (standard PC ratio)

If the ratio changes, the image becomes **stretched or compressed**.

4. Refreshing

In **CRT displays**, phosphor dots glow for a short time and then fade.So the electron beam redraws the image many times per second.This process is called **refreshing**.

Normal refresh rate:60 to 80 frames per second

Digital Image and Pixels

A **digital image** is made of many very small dots called **pixels (picture elements)**. These pixels are arranged in **rows and columns** to form a rectangular picture. This arrangement is called a **raster**.

Each pixel represents a small part of the image. When many pixels are placed together, they form the complete picture that we see on a screen or print.

The **total number of pixels** in an image depends on two things:

1. The **size of the image**
2. The **number of pixels per unit length** (for example pixels per inch)

The number of pixels present in **one inch of an image** is called the **resolution**. Higher resolution means more pixels are used, so the image becomes **clearer and more detailed**.

For example, if an image is **3 × 2 inches** and the resolution is **300 pixels per inch**, then the total pixels will be:

- Horizontal pixels = 3 × 300 = 900
- Vertical pixels = 2 × 300 = 600
- Total pixels = 900 × 600 = **540,000 pixels**

Sometimes image size is written as **number of pixels in horizontal direction × number of pixels in vertical direction**, such as **640 × 480** or **1024 × 768**. However, this only tells the **pixel count**, not the actual physical size or resolution of the image.

For example, a **640 × 480 image** can appear larger or smaller depending on the resolution used when displaying or printing it.

Aspect Ratio

The **aspect ratio** of an image is the **relationship between its width and height**.

It tells us how wide the image is compared to its height.

For example:

- A **2 × 2 inch image** has an aspect ratio of **1 : 1** because width and height are equal.
- A **1024 × 768 image** has an aspect ratio of **4 : 3**.

Images with the same aspect ratio keep the **same shape**, even if the size changes.

Pixel Coordinates

Every pixel in an image can be identified by **coordinates**. A coordinate system is used to locate pixels.Usually, the **lower-left corner** of the image is taken as the **origin (0,0)**.Example for a 640 × 480 image:

- Lower-left pixel → **(0,0)**
- Lower-right pixel → **(639,0)**
- Upper-right pixel → **(639,479)**

These coordinates help the computer know **exactly where each pixel is located**.

Image Creation in Computer Graphics

Creating an image on a computer mainly means **assigning color values to pixels**. Every pixel gets a specific color value. When millions of pixels are given different colors, together they form the picture that we see.

RGB Color Model

The **RGB color model** is the most commonly used method to represent colors in computer graphics.

It uses three primary colors:

- **Red (R)**
- **Green (G)**
- **Blue (B)**

Each of these colors can have an **intensity value from 0 to 1**:

- **0** → color is off
- **1** → color is fully on

By mixing different amounts of these three colors, many different colors can be created. All these colors together form the **RGB color space**, which is represented as a cube.Examples of RGB values:

- **(0,0,0)** → Black
- **(1,1,1)** → White
- **(1,1,0)** → Yellow
- **(0.7,0.7,0.7)** → Gray

The diagonal line between black and white in the cube is called the **gray axis**, which contains different shades of gray.

Working Principle of RGB Model

- Start with **black color**.
- Add **red, green, and blue light** in different intensities.
- Mixing these lights produces different colors.
- When all three are fully added, the result is **white**.

This method is called an **additive color process**, and it matches how **display monitors work**.

CMY Color Model

The **CMY color model** is another color system mainly used in **printers**.

It uses three colors:

- **Cyan (C)**
- **Magenta (M)**
- **Yellow (Y)**

This model works differently from RGB. Instead of adding colors, it **removes colors from white**.

Working Principle of CMY Model

- Start with **white color**.
- Subtract certain colors from white.
- The remaining colors form the final color.

For example:If **red is removed from white**, the remaining colors are **green and blue**, which appear as **cyan**.This process is called the **subtractive color process**, and it matches how **printers produce colors**.

Direct Coding (Image Representation)

In **direct coding**, each pixel directly stores the color information. A fixed number of **bits of memory** is assigned to each pixel to represent its color.

For example, a **3-bit color representation** may use:

- 1 bit for **Red**
- 1 bit for **Green**
- 1 bit for **Blue**

This allows only **8 possible colors**.

Example Colors in 3-bit Coding

- 000 → Black
- 001 → Blue
- 010 → Green
- 011 → Cyan

- 100 → Red
- 101 → Magenta
- 110 → Yellow
- 111 → White

True Color Representation

A common standard is **24-bit color**, where:

- **8 bits for Red**
- **8 bits for Green**
- **8 bits for Blue**

This allows:

- **256 intensity levels per color**
- Total colors = **256 × 256 × 256 = 16.7 million colors**

This is called **true color**, because the color differences become almost impossible for the human eye to notice.

Special Cases

Black-and-white image

- Uses **1 bit per pixel**
- 0 = black
- 1 = white

Grayscale image

- Usually uses **8 bits per pixel**
- Allows **256 shades of gray**

Lookup Table Method

The **lookup table method** reduces the storage space needed for images.

In this method, the **pixel values do not directly store colors**. Instead, the pixel value acts as an **index (address)** that points to a color stored in a table.

The table contains the actual **RGB color values**.

For example:A lookup table may contain **256 entries.**Each entry stores a **24-bit RGB color**Pixels only store **8-bit values**, which point to one of those 256 colors.So the pixel value tells the system which color in the table should be used.This method is often called the **8-bit image format**.

Advantages of Lookup Table Method

- Requires **less storage space**
- Allows **efficient color management**
- Faster for some graphics operations

Disadvantages of Lookup Table Method

- Only **limited number of colors** can appear at the same time (e.g., 256)
- Changing the color table can **change the appearance of the whole image**

Display Monitor

A **display monitor (video monitor)** is an output device that converts **digital images from a computer into visible pictures** on the screen.

One of the earliest and most common display technologies is the **CRT (Cathode Ray Tube)** monitor. A CRT monitor is basically a **vacuum glass tube** that produces images by using **electron beams** and a special light-producing material called **phosphor**.

Inside the display screen of the CRT there is a **phosphor coating**. When this phosphor is hit by an electron beam, it **produces light**. Because of this, small bright spots appear on the screen. When many such spots are produced quickly across the screen, they form the complete image that we see.

The light produced during the electron beam hitting the phosphor is called **fluorescence**. Even after the beam moves away, the phosphor continues glowing for a short time. This continued glow is called **phosphorescence**, and the duration of this glow is known as **persistence**.

Main Parts of a CRT Monitor

A CRT monitor mainly consists of:High voltage (around **15,000–20,000 volts**) accelerates the electrons toward the screen.

- Electron gun-Produces a continuous stream of electrons.
- Control electrode-Controls the intensity of the electron beam.
- Focusing electrode-Focuses electrons into a **narrow beam**.
- Horizontal deflection plates-Directs the beam to different positions on the screen.
- Vertical deflection plates-Directs the beam to different positions on the screen.

- Phosphor-coated screen-Produces light when struck by electrons.
- Control circuits
- **High-voltage system** – Accelerates electrons toward the screen.

Working Principle of a Monochrome CRT Monitor

- The **electron gun** produces a stream of electrons by heating a metal surface.
- The **control electrode** controls the number of electrons in the beam.
- The **focusing electrode** focuses the electrons into a narrow beam.
- The electron beam travels toward the **phosphor-coated screen**.
- When the beam hits the phosphor, it produces **light spots** on the screen.
- **Horizontal deflection plates** move the beam from **left to right** across the screen.
- **Vertical deflection plates** move the beam from **top to bottom** of the screen.
- The beam scans the screen **line by line**, lighting up each pixel.
- After completing a line, the beam quickly returns to the start of the next line (called **horizontal retrace**).
- After finishing the whole screen, it returns to the top again (**vertical retrace**).

By repeating this scanning process very quickly, the monitor displays a **complete image**.

Frame Buffer and Refresh Rate

The image shown on the monitor is stored in a special memory called the **frame buffer (or refresh buffer)**. This memory stores the **pixel values** of the image.

The monitor continuously reads data from the frame buffer and displays it on the screen.

The **refresh rate** is the number of times the screen is updated every second.

Typical refresh rate:

- **60 frames per second (60 Hz)** or higher.

If the refresh rate is too low, the image may appear to **flicker**, which is uncomfortable for the human eye.

The persistence of phosphor must be balanced:

- **Long enough** so the image stays visible
- **Short enough** so the next frame can appear clearly

Interlacing Technique

Some monitors use a method called **interlacing** to reduce flickering.

Working Principle of Interlacing

- The screen is divided into **two sets of scan lines**.
- First, the monitor refreshes **all odd-numbered lines**.
- Then it refreshes **all even-numbered lines**.
- This makes the screen appear to refresh faster and reduces flicker.

Although the full screen refresh rate is the same, the technique **improves visual stability**.

Color Display (Color CRT Monitor)

A **color monitor** works similarly to a monochrome CRT but it can display many colors.

Instead of one electron gun, a color CRT monitor has **three electron guns**:

- One for **Red**
- One for **Green**
- One for **Blue**

The inside of the screen contains **tiny dots of three different phosphor materials** that produce **red, green, and blue light**.

When the electron beams hit these phosphor dots, they produce colored light. When viewed from a normal distance, the red, green, and blue lights mix together and create **many different colors**.

A thin metal plate called a **shadow mask** is placed between the electron guns and the phosphor screen. The shadow mask has very small holes that ensure each electron beam hits the **correct phosphor dot**.

The distance between these phosphor dots is called the **dot pitch**. A smaller dot pitch generally means **better image sharpness and higher resolution**.

Techniques for Producing Color in CRT

1. Beam Penetration Method

2. Shadow Mask Method

1. Beam Penetration Method

This method is mainly used in **random scan monitors**.

Working Principle

- The screen has **two layers of phosphor**.
- The **outer layer is red**, and the **inner layer is green**.
- The color depends on **how deep the electron beam penetrates** into the phosphor layers.
- By controlling the beam energy, different colors are produced.

Colors Produced-Red,Orange,Yellow,Green

Advantages

- **Low cost**
- Simple design

Disadvantages

- Produces **limited number of colors**
- **Poor picture quality**

2. Shadow Mask Method

This method is commonly used in **raster scan CRT monitors**.

Working Principle

- The screen has **three phosphor dots** at each pixel: **Red,Green,Blue**
- The CRT contains **three electron guns**, one for each color.
- A **metal plate called a shadow mask** is placed in front of the phosphor screen.
- The shadow mask has **tiny holes** that guide the electron beams.
- Each beam hits its **correct color phosphor dot**.
- The lights from the three dots mix together to produce **different colors**.

Advantages

- Produces **wide range of colors**
- **Better image quality**

Disadvantages

- **More expensive**
- **More complex structure**

1.**Random Scan Display**

Random scan displays are also called:

- **Stroke-writing displays**
- **Vector displays**
- **Calligraphic displays**

Working Principle

- The electron beam moves **only to the parts of the screen where the picture needs to be drawn**.
- It draws images **line by line directly** instead of scanning the entire screen.

Features

- Very **good for line drawings**
- Produces **smooth lines**

Limitations

- Cannot display **realistic images or shaded pictures**

2.**Raster Scan Display**

Raster scan displays are based on **television technology**.

Working Principle

- The electron beam scans the screen **one line at a time from top to bottom**.
- Each line is called a **scan line**.
- The beam moves **left to right**, then quickly returns to the start of the next line (**horizontal retrace**).
- After finishing the entire screen, it returns to the top (**vertical retrace**).
- The intensity of the beam is controlled according to the **image data stored in the frame buffer**.

Refresh Rate

- The screen is refreshed **60–80 times per second**.

3.Direct View Storage Tube (DVST)

DVST is another type of CRT display that **does not require continuous refreshing**.

In this system, the image is stored as a **charge pattern behind the phosphor screen**.

Two electron guns are used:

- **Primary gun** – Stores the picture pattern.
- **Flood gun** – Maintains the image on the screen.

DVST vs Refresh CRT

Advantages of DVST

- No need for **continuous refreshing**
- Can display **complex images with high resolution**
- **No flickering**

Disadvantages of DVST

- Usually **cannot display colors**
- **Parts of the image cannot be erased**
- Not suitable for **animation**

Advantages of CRT Display

- Provides **high resolution**
- **Wide viewing angle**
- **Cheaper** compared to LCD and plasma displays

Disadvantages of CRT Display

- **Large and bulky**
- **Heavy and fragile**
- Not suitable for **small portable devices**
- Display area is **smaller than the monitor size**

Flat Panel Displays

A **Flat Panel Display (FPD)** is a modern display technology that is **thinner, lighter, and consumes less power** than the traditional **CRT monitor**.

Unlike CRT displays, flat panel displays do not use a large glass tube. Because of their **small size and low power consumption**, they are widely used in **modern electronic devices** such as laptops, televisions, mobile phones, and calculators.

Flat panel displays are designed to save **space, weight, and electricity**, which makes them suitable for **portable and battery-powered devices**.

Types of Flat Panel Displays

There are two main types of flat panel displays:

1. Emissive Displays

2. Non-Emissive Displays

1. Emissive Displays

Emissive displays are devices that **produce their own light**. They convert **electrical energy directly into light** to form images.

Examples include:Plasma Display Panel (PDP),Light Emitting Diodes (LED)

Because they produce their own light, they **do not need an external light source**.

2. Non-Emissive Displays

Non-emissive displays **do not produce their own light**. Instead, they use **external light** (like sunlight or a backlight) and control how the light passes through the display to create images.

Example:Liquid Crystal Display (LCD)

These displays work by **modifying or blocking light** to form graphics and images.

Liquid Crystal Display (LCD)

A **Liquid Crystal Display (LCD)** is a **thin and flat display device** made up of many small picture elements called **pixels**. These pixels are arranged in rows and columns to form the image.

LCDs use **very little electrical power**, so they are ideal for **battery-powered devices** like watches, calculators, laptops, and mobile phones.Color images can be produced in LCDs by using **color filters** for red, green, and blue.

The main component of an LCD is the **liquid crystal material**. This material behaves partly like a **liquid** and partly like a **solid crystal**.

- Like a liquid, it can **flow**.
- Like a crystal, its **molecules are arranged in an ordered structure**.

Because of this property, liquid crystals can **change their orientation when an electric field is applied**, which helps control the light passing through the display.

LCD Construction

Each pixel in an LCD contains several layers that control the light.

Basic Structure (Layers of LCD)

- Reflective Layer
- Horizontal polariser
- Horizontal Grid Wires
- Liquid crystal layer
- Vertical Grid Layer
- Vertical Polarizer

The **liquid crystal molecules are placed between two transparent electrodes**. Two polarizing filters are placed on both sides of the liquid crystal layer.These layers work together to control the light and create the image.

Basic Principle of LCD

Most LCDs use a structure called **Twisted Nematic (TN)** liquid crystal.

In this structure, the liquid crystal molecules form a **twisted or helical arrangement**. This twist helps rotate the **polarization of light** passing through the display.

When an electric field is applied, the molecules change their orientation. This change affects how light passes through the display and allows pixels to appear **bright or dark**.

Working Principle of LCD

- Liquid crystal molecules normally form a **twisted (helical) structure**.
- Light passing through the first polarizing filter becomes **polarized**.
- The twisted liquid crystal molecules **rotate the polarization of light**.
- The rotated light passes through the second polarizing filter and becomes **visible**.
- When an **electric voltage is applied**, the molecules align with the electric field.
- This alignment **removes the twist** in the structure.

- As a result, the light is **blocked by the second polarizer**.
- By controlling the voltage at each pixel, the display creates **images and patterns**.

Classification of LCD

1. Transmissive LCD

2. Reflective LCD

1. Transmissive LCD

- Uses a **backlight source** behind the display.
- Light passes **through the LCD panel** and is viewed from the front.
- Commonly used in **computer monitors, laptops, and TVs**.

2. Reflective LCD

- Uses **external light** instead of a backlight.
- Light is **reflected from a reflector behind the display**.
- Commonly used in **calculators and digital watches**.

Applications of LCD

- Digital watches,Pocket calculators,Mobile phones
- Laptop screens,Computer monitors,Televisions,Personal organizers

Advantages of LCD

- **Very low power consumption**
- **Thin and lightweight**
- Works at **low voltage (2–3 V)**
- Suitable for **battery-powered devices**

Limitations of LCD

- Cannot be seen clearly **in complete darkness** without backlight
- **Limited viewing angle**
- Works only in a **limited temperature range**

Thin Film Transistor (TFT) Technology

In simple LCD displays such as those used in **calculators**, the pixels are **directly controlled by voltage**. This means a voltage is applied to a segment to turn it on or off.

However, this method becomes **impractical for large displays** like computer monitors or televisions because these displays contain **millions of pixels**. If every pixel needed a direct connection for **red, green, and blue**, the display would require **millions of electrical connections**, which is not practical.

To solve this problem, the pixels are arranged in **rows and columns**. This greatly reduces the number of electrical connections. But for better control of each pixel, **each pixel is given its own transistor switch**.

This technology is called **Thin Film Transistor (TFT)** technology. TFT allows every pixel to be controlled **individually**, which improves **image quality, brightness, and response time**.

A **TFT is a special type of field-effect transistor** created by depositing **very thin layers of semiconductor and metal materials**. When TFT technology is used with LCD displays, it forms **TFT-LCD**, which is a type of **active matrix LCD**.

TFT-LCD displays are widely used in **modern monitors, laptops, televisions, smartphones, and projectors**.

Construction of TFT Displays

Materials Used

- **Metal–Insulator–Semiconductor Field Effect Transistors (MISFETs)** are commonly used.
- **Indium Tin Oxide (ITO)** is used for transparent electrodes.
- Many TFTs are made using **amorphous silicon (A-Si:H)**.
- These materials help reduce the **number of manufacturing steps** and make production easier.

Construction

- Each pixel contains **a transistor switch**.
- This transistor allows **individual control of the pixel**.
- Every pixel acts like a **small capacitor**.
- The pixel consists of:
  - A transparent **ITO layer at the front**
  - A transparent **ITO layer at the back**

    - A **liquid crystal layer** between them

- All pixels in a **row receive positive voltage**.
- All pixels in a **column receive negative voltage**.

This arrangement allows **precise control of each pixel**, resulting in clearer images.

Advantages of TFT Displays

- **Fast response time**
- **Lower production cost**
- **Less noise or glitter on the screen**
- Produces **better image quality than traditional LCDs**

Disadvantages of TFT Displays

- **Limited viewing angle**, especially vertically
- Cannot always display the **full 16.7 million colors** available from some modern graphics systems

Plasma Display Panel (PDP)

A **Plasma Display Panel (PDP)** is a type of **flat panel display** that is commonly used for **large television screens**.

Plasma displays use **small cells filled with gas** instead of liquid crystals. When electricity passes through the gas, it turns into **plasma**, which produces **light**.

These displays are usually used for **large screens (above 37 inches)** and can be made very large.

Construction of Plasma Display

- The display contains **hundreds of thousands of tiny cells**.
- These cells contain **xenon and neon gases**.
- The cells are placed between **two glass plates**.
- **Electrodes** are placed in front and behind the cells.
- Each cell is protected with a **magnesium oxide layer**.

Working Principle of Plasma Display

- Control circuits send electrical signals to the **electrodes**.

- When two electrodes intersect at a cell, a **voltage difference is created**.
- This voltage causes the **gas inside the cell to ionize and form plasma**.
- The plasma releases **ultraviolet photons**.
- These photons hit the **phosphor coating inside the cell**.
- The phosphor then produces **visible colored light**.

This process forms the image on the screen.

Features of Plasma Displays

- Panel thickness is about **6 cm**.
- Total thickness is usually **less than 10 cm**.
- Screen sizes can reach up to **103 inches**.
- Typical lifetime is around **60,000 hours**.

Advantages of Plasma Displays

- Supports **very large display sizes**
- **Thin design** and can be mounted on walls
- **Fast response time**
- **Wide viewing angles**
- Produces **rich colors and high brightness**

Disadvantages of Plasma Displays

- **More expensive** than CRT and LCD displays
- **Burn-in problem**
  Static images can leave a **permanent mark on the screen**
- Because of this issue, they are **not suitable for computer monitors or video games**

1) TFT-LCD vs Normal LCD (Direct Driven LCD)

Feature	Normal LCD	TFT-LCD (Active Matrix)
Pixel Control	Directly driven (voltage applied to segment)	Each pixel has its own **transistor switch**
Image Quality	Lower	Higher, sharper images
Size Limit	Small displays (calculators, watches)	Works for **large displays** (TV,

Speed	Slower response time	Fast response time
Number of Connections	Millions of wires needed	Reduced connections (rows & columns)
Cost	Cheaper for small displays	Slightly higher cost for

Advantages & Disadvantages

- **Normal LCD**
    - Advantages: Simple, cheap for small displays
    - Disadvantages: Cannot be used for large screens, slower response
- **TFT-LCD**
    - Advantages: High image quality, fast response, precise pixel control
    - Disadvantages: Limited viewing angle, cannot always display full 16.7 million colors

2) LCD vs LED (Light Emitting Diode)

Feature	LCD	LED
Backlight	Uses **fluorescent tubes**	Uses **LED backlight**
Thickness	Thicker	Thinner and lighter
Power Consumption	More power	Less power
Brightness	Moderate	Brighter images
Color & Contrast	Moderate	Better color and contrast
Lifespan	Shorter	Longer lifespan
Cost	Cheaper	Slightly more expensive

Advantages & Disadvantages

- **LCD**
    - Advantages: Cheap, widely available
    - Disadvantages: Less bright, thicker, consumes more power
- **LED**
    - Advantages: Thin, bright, energy efficient, better colors
    - Disadvantages: Slightly expensive

3) QLED vs OLED

Feature	QLED (Quantum Dot LED)	OLED (Organic LED)
Light Source	LED backlight + quantum dots	Each pixel emits its own light
Black Levels	Good, but not perfect	Perfect black (pixel can turn off completely)
Brightness	Very high	Moderate brightness
Color	Excellent	Excellent
Viewing Angle	Narrower	Wider viewing angle
Thickness	Thicker (needs backlight)	Very thin (no backlight)
Lifespan	Long	Slightly shorter (burn-in possible)
Cost	Cheaper than OLED	More expensive

Advantages & Disadvantages

- **QLED**
    - Advantages: Very bright, long-lasting, better for daylight
    - Disadvantages: Blacks not perfect, thicker, narrower viewing angle
- **OLED**
    - Advantages: Perfect blacks, wide viewing angle, thin design
    - Disadvantages: Burn-in problem with static images, expensive
- Think **TFT-LCD = each pixel has transistor**
- **LED = LCD with LED backlight**
- **QLED = LED + quantum dots**
- **OLED = each pixel lights itself**

TFT Display vs Plasma Display

Feature	TFT Display (Thin Film Transistor LCD)	Plasma Display (PDP)
Basic Technology	Uses **liquid crystal technology with thin film transistors** to control pixels	Uses **tiny gas-filled cells (neon and xenon)** that form plasma to produce light
Construction	Each pixel contains a **thin film transistor (TFT) switch** and a **liquid crystal layer between two glass plates**	Consists of **two glass plates with thousands of tiny cells filled with neon and xenon gas**

Pixel Structure	Each pixel acts like a **small capacitor with a transistor for individual control**	Each pixel contains **three small gas cells with phosphor coating for RGB colors**
Materials Used	Uses **liquid crystals, transparent electrodes (Indium Tin Oxide), and thin film transistors**	Uses **gas (neon & xenon), phosphor coating, electrodes, and protective magnesium oxide layer**
Light Source	Needs a **backlight** because liquid crystals do not produce light	**Self-emitting** display (plasma produces light directly)
Working Principle	Voltage applied to TFT **controls orientation of liquid crystal molecules**, which controls the amount of light passing through the pixel	Voltage applied to electrodes **ionizes the gas**, forming plasma which emits **UV light that excites phosphor to produce visible light**
Pixel Control	Each pixel is **individually controlled by its own transistor**	Pixels are controlled by **electrodes crossing each gas cell**
Image Formation	Image is formed by **controlling light passing through liquid crystal pixels**	Image is formed by **light emitted from plasma-excited phosphor cells**
Power Consumption	**Lower power consumption**	**Higher power consumption**
Screen Size	Common for **monitors, laptops, tablets**	Mostly used for **large TVs**
Thickness	**Very thin and lightweight**	Thin but **slightly thicker than TFT**

- **TFT:** Liquid crystal + transistor control + backlight
- **Plasma:** Gas cells + plasma + phosphor light

Display Type	**Technology Used**	**Light Source**	**Screen Thick**	**Image Quality**	**Power Consumption**	**Advantages**	**Disadvantages**
CRT (Cathode	Electron beam hits phosphor	Self-emitting	Very thick	Good color and	High	Wide viewing angle, good	Very heavy, bulky,
LCD (Liquid	Liquid crystals control light	External backlight	Thin	Moderate quality	Low	Lightweight, low power, used	Limited viewing angle,
TFT-LCD	LCD with **Thin Film Transistor for**	Backlight	Very thin	Better image quality	Low	Fast response, better image control, sharper	Limited viewing angle, slightly costly
LED Display	LCD with LED backlight	LED backlight	Very thin	Bright and good	Very low	Energy efficient, long life, thin	Costlier than normal LCD
QLED	LED with **quantum dot**	LED backlight	Thin	Very bright and	Moderate	Very high brightness, long	Black levels not perfect
OLED	Organic materials emit	Self-emitting	Extremely	Excellent color and	Low	Perfect blacks, wide viewing	Expensive, burn-in
Plasma Display	Gas cells become plasma	Self-emitting	Thin	Very bright	High	Large screens, wide viewing	High power use, burn-in